Put $x_3 = \alpha$, $\alpha \in R$

$X = \alpha(-1, -1, 1)$

$N(A) = \{\alpha(-1, -1, 1)\}$, $\alpha \in R$

System of Linear equation

$AX = b$

Consistent :-

A system of Linear equation $AX = b$, $A = [a_{ij}]_{m \times n}$, $b = (b_1, b_2, \ldots, b_m)^T$, $X = (x_1, x_2, \ldots, x_n)^T$ is said to be consistent (solvable) if $\boxed{\text{rank}(A) = \text{rank}(C)}$, where $C = [A : b]$ ↓ Augmented matrix, otherwise system will be inconsistent.

Homogeneous System ($AX = 0$)	Solution
1) $\text{rank}(A) = \text{rank}(C) = r = m$ $(m < n)$	Infinite solution
2) $\text{rank}(A) = \text{rank}(C) = r = m = n$	Trivial solution
3) $\text{rank}(A) = \text{rank}(C) = r = n < m$	unique solution

Non-homogeneous System ($AX = b$)	Solution
1) $\text{rank}(A) = \text{rank}(C) = r = m < n$	Infinite solution
2) $\text{rank}(A) = \text{rank}(C) = r = m = n$	Unique solution
3) $\text{rank}(A) = \text{rank}(C) = r = n < m$	Unique solution
4) $\text{rank}(A) \neq \text{rank}(C)$	No solution

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